

Measuring Validator and Sandboxed-Execution Coverage for LLM-Generated NetOps Artifacts: A Paired Confirmatory Study

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Abstract—We preregistered and executed a paired confirmatory experiment (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e) that measured the marginal detection benefit of adding sandboxed emulation to a deterministic validator for LLM-generated intent→configuration artifacts. Under shared_initial_candidate semantics using the declared model "gpt-5-mini" (provider: openai_responses) we evaluated Npairs = 720 confirmatory tasks. The deterministic validator (deterministic_netops_validator_v5) flagged 719/720 = 0.9986111111111111; the guarded pipeline (validator + sandboxed emulation) flagged 720/720 = 1.0. Paired difference (guarded - baseline) = 0.001388888888888884; exact McNemar two-sided p = 1.0; 95% bootstrap percentile CI (10000 resamples, seed = 1729) = [0.0, 0.004166666666666667]. Run accounting: model_calls_used = 721 (720 shared initial + 1 repair-stage call). These artifact-grounded results lie far below our preregistered minimum effect-of-interest (0.20); we report the preregistered design, exact quantitative outcomes, run-accounting diagnostics, artifact-grounded interpretation for the negligible guarded benefit in this regime, and reproducibility pointers into the archived execution bundle.

I. INTRODUCTION

Generative large language models (LLMs) are increasingly proposed to translate operator intent into device configuration and NetOps workflows. Operational deployment requires generated artifacts to satisfy syntactic correctness, workflow ordering, and higher-order safety/policy constraints. A common mitigation architecture is generate→validate→(repair)→execute, sometimes augmented by sandboxed emulation to reveal runtime-only failures that static validators miss. Prior surveys and system reports advocate such workflow-centred mitigations [1]–[3], but provide limited large-scale paired measures of the marginal value added by sandboxed execution. Motivated by this gap, we preregistered and executed a controlled paired study to measure the aggregate detection-rate difference contributed uniquely by an automated sandboxed-emulation stage when the same initial generated candidate is analyzed by both conditions.

Research question and contributions. Our preregistered research question was: for synthetic intent→configuration tasks under shared_initial_candidate semantics, what is the paired detection-rate difference (guarded - baseline) between (A) a deterministic validator-only pipeline and (B) the same deterministic validator followed by automated sandboxed emulation? Contributions: (1) a machine-readable preregistration and faithful autonomous execution; (2) an artifact-grounded

paired analysis on Npairs = 720 with complete accounting; (3) an artifact-grounded interpretation explaining a near-zero marginal effect; (4) concise reproducibility pointers into the archived bundle and prescriptive boundary conditions where guarded stages are likely to matter.

II. RELATED WORK

Workflow-centred mitigations and verification tools. Surveys and recent system reports document architectures that pair LLM generation with validators, verifiers, and emulators to improve correctness of generated NetOps artifacts and adjacent program-generation tasks [1], [2]. Tool-focused work demonstrates generate→validate→repair patterns (e.g., Batfish/NetKAT style verification, emulator-assisted validation) but reports heterogeneous gains depending on task, validator rule coverage, and emulator fidelity [2], [3].

Empirical findings motivating a paired design. Empirical studies in program/specification generation show external verifiers often catch model mistakes not found by the model alone, and some NetOps prototypes use emulation to expose runtime issues [3]. These prior findings motivated our preregistered paired design that isolates guarded-stage contributions: because we enforce shared_initial_candidate semantics, the only permissible sources of between-condition discordance are downstream guarded-stage emulation observations and permitted repair-stage invocations recorded in run accounting.

III. METHODOLOGY

Preregistration and analytic intent. We preregistered a confirmatory paired-binary design (study_id f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e). The primary estimand was the paired_success_rate_difference_guarded_minus_baseline aggregated across confirmatory samples; the preregistered minimum effect-of-interest for power planning was 0.20 absolute. The preregistration (frozen prior to execution) specified: inclusion/exclusion rules, deterministic missingness and reseed policies (up to two automatic reseeds for parser failures), repair-stage limits (≤ 1 repair call per task), contamination detection and deterministic replacement rules, and the analysis executor (paired_binary_analysis_v1) with paired inference procedures and pre-registered sensitivity analyses.

Task bank, topology templates, and vendor abstraction. Confirmatory items were synthetic intent→configuration tasks generated from the intent_configuration_repair_v1 family. The

bank covered three abstracted vendor-CLI families and three topology templates (leaf-spine, 3-node service chain, simple triangle). Each task manifest encoded: the initial device state, an explicit intent, a required command sequence, allowed-touched fields, and parser-acceptance constraints. Inclusion required vendor-specific parser acceptance after up to two deterministic reseeds; irrecoverable parser failures were to be deterministically excluded and replaced per the preregistered missingness plan. In the executed run there were zero irrecoverable parser exclusions and `complete_pair_count = 720` (analysis/missingness_summary.csv).

Model interface, sampling semantics, and shared_initial_candidate enforcement. The declared/requested model was recorded as "gpt-5-mini" and the execution model_configuration.json contains `declared_model_version = "gpt-5-mini"`, `max_output_tokens = 2000`, and `temperature = null` (sha256 a40e96e212ab87da...). Important operational clarification (preregistered and enforced): `temperature = null` in the archived configuration indicates the client did not set an explicit temperature parameter; null does not imply `temperature=0` or provider-side determinism. To isolate guarded-stage effects we enforced shared_initial_candidate semantics: for each pair the pipeline performed a single sampled initial generation (shared) which both the baseline validator and the guarded pipeline consumed. Concretely, provider_call_audit and deterministic_reconciliation artifacts record stage_counts: `shared_initial_generation = 720` and `repair = 1`, and `model_calls_used = 721` (720 shared + one repair-stage call). Because the same sampled candidate was analyzed by both conditions for each pair, any between-condition discordance must stem from (i) guarded-stage sandboxed emulation observations not encoded as validator violations, and/or (ii) the permitted repair-stage intervention recorded in run accounting.

Validator implementation and scoring rule. Baseline validation used deterministic_netops_validator_v5 (scorer_version = 5.0) executing the preregistered scoring rule full_intent_constraint_validation. The validator performs multi-level checks recorded in per-episode validator_trace objects: vendor-specific parsing, syntactic command parsing, command ordering/workflow checks against required_sequence, required-field presence, dependency constraints (e.g., must-shutdown-before-mtu-change patterns), and policy constraints (protected interfaces, group-wise minimum-active constraints). The validator emits structured outputs used in scoring: `normalized_configuration`, `parsed_command_count`, `required_command_count`, `observed_touched_field_count`, `violation_count`, and `violation_codes`. Per-episode artifacts (execution/scoring/task-XXXXX-*.json) show these fields and `validator_version = 5.0` for all confirmatory episodes.

Sandboxed-emulation harness and guarded pipeline semantics. The guarded pipeline applied the baseline deterministic validator and then, for candidates not blocked by parser failure, executed an automated sandboxed-emulation harness that instantiated an emulated representation of the

declared topology and applied the candidate commands. The emulation harness produced runtime observations that were normalized and summarized into the same validator-trace structure (`validation_before/validation_after`, `final_state`, `parsed_command_count`, `observed_touched_field_count`). The emulation stage was designed to detect runtime-only symptoms such as transient ordering-dependent state changes, reachability/regression signals, or state transitions that static rules do not explicitly capture. The preregistered repair policy permitted up to one repair-stage model call per task; repair calls (if used) and their provider traces are recorded in the execution manifest and deterministic_reconciliation artifacts.

Inclusion/exclusion, missingness, and contamination controls. The preregistered missingness plan allowed two automatic deterministic reseeds for parser failures; irrecoverable parser failures would be deterministically replaced. Contamination controls included deterministic SHA-256 artifact hashing, automated prompt-template divergence checks, and validator/emulator binary/config checksum verification before confirmatory execution. The execution manifest and analysis/contamination_summary.csv report no contamination flags and no irrecoverable parser exclusions for the confirmatory set.

Statistical analysis procedures. The primary confirmatory analysis used the paired_binary_analysis_v1 executor specified in the preregistration. The primary test computed the aggregate paired detection-rate difference (guarded - baseline) across complete pairs and reported an exact McNemar two-sided p-value. Pre-registered uncertainty quantification used paired-bootstrap percentile confidence intervals with 10,000 resamples and a fixed seed (`bootstrap_seed = 1729`) for reproducibility. Secondary exploratory analyses (per-class and per-vendor paired differences) were pre-registered with Bonferroni adjustments; these are archived but not the focus of the confirmatory report. The preregistered default treatment of irrecoverable technical failures for the primary analysis was to exclude complete-pair technical losses (`complete_pair_primary`); a sensitivity analysis treats irrecoverable failures as nondetections (documented in the preregistration). The executed analysis followed the preregistered `complete_pair_primary` plan; analysis artifacts and logs are archived (analysis/analysis_log.jsonl and analysis/paired_contingency_table.csv).

Deterministic reconciliation and run accounting. To enable artifact-grounded claims and independent checking we recorded provider_call_audit and deterministic_reconciliation artifacts. These reconcile: `planned_episode_count = 1440`, `completed_episode_count = 1440`, `complete_pair_count = 720`, `failed_episode_count = 0`, `model_calls_used = 721`, and stage_counts (`shared_initial_generation = 720`; `repair = 1`). These machine-verifiable run-accounting artifacts are referenced throughout the manuscript when we state numerical outcomes and when tracing the single discordant guarded-only detection back to a documented repair-stage invocation.

IV. RESULTS

Primary confirmatory counts and test statistics (artifact-grounded). Analysis artifacts (analysis/paired_contingency_table.csv and analysis/condition_summary.csv) report the contingency counts: $n_{11} = 719$ (detected by both), $n_{10} = 1$ (guarded-only), $n_{01} = 0$ (baseline-only), $n_{00} = 0$ (neither). Marginal detection rates: baseline = $719/720 = 0.9986111111111111$; guarded = $720/720 = 1.0$. Paired estimate (guarded - baseline) = 0.0013888888888888884 . Exact McNemar two-sided test $p = 1.0$. Paired-bootstrap percentile 95% CI (10000 resamples, bootstrap_seed = 1729) = $[0.0, 0.004166666666666667]$. All numeric values below are reproduced verbatim from analysis/results.json and analysis artifacts.

Table 1 — Primary confirmatory summary (artifact-grounded). - Npairs = 720 (complete paired episodes) - Baseline successes = 719; Baseline success rate = 0.9986111111111111 - Guarded successes = 720; Guarded success rate = 1.0 - Paired difference (guarded - baseline) = 0.0013888888888888884 - Exact McNemar two-sided $p = 1.0$ - 95% bootstrap percentile CI = $[0.0, 0.004166666666666667]$ (10000 resamples, seed = 1729)

Discordant-pair attribution and repair accounting. Provider-call accounting and deterministic reconciliation identify a single repair-stage invocation (stage_counts: repair = 1) and a unique n_{10} discordant count. Because shared_initial_candidate semantics were enforced, the permissible causes for that single guarded-only detection are: (i) a guarded-stage emulation observation not encoded as a validator violation, or (ii) the single recorded repair-stage invocation. The confirmatory execution bundle contains the raw per-episode records; the unique discordant record can be located by searching execution/raw_results.jsonl (input-results SHA-256 = `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`) for the single entry where baseline_score = 0 and guarded_score = 1. Deterministic reconciliation is archived as analysis/deterministic_reconciliation.json (sha256 `49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa`) and its provider_call_audit confirms model_calls_used = 721 and repair = 1. The raw_results record for the discordant entry contains the pair_id and the validator_trace and repair_provider_trace fields that link the guarded detection to the documented repair-stage call; these artifacts are part of the canonical execution bundle referenced by the execution manifest.

Representative per-episode diagnostics. Representative scoring traces archived under execution/scoring/task-000001-*.json ... task-000006-*.json include validator_trace objects with normalized_configuration, validation_before/after, parsed_command_count, required_command_count, violation_count, and validator_version = 5.0. Those representative artifacts illustrate that the vast majority of shared_initial candidates complied with validator rules (violation_count = 0) and that sandboxed emulation largely corroborated validator findings in this curated task bank.

V. DISCUSSION

Why was the guarded-stage aggregate benefit negligible? The archived evidence supports three artifact-grounded factors.

1) Validator ceiling effect. Deterministic_netops_validator_v5 covered the syntactic and explicit semantic properties encoded in the synthetic task bank (shutdown-configure-restore, make-before-break uplink switchover, paired MTU changes). Baseline success rate ≈ 0.9986 indicates near-universal validator coverage for these curated patterns, leaving negligible headroom for the guarded stage.

2) Task curation and inclusion policy. Confirmatory items were parser-validated synthetic tasks specifically designed to exercise workflow policies but not constructed adversarially to evade validator checks. This curation concentrates cases inside the validator’s coverage envelope and reduces the prevalence of subtle runtime-only failure modes that emulation might reveal.

3) Shared_initial_candidate semantics and limited repair activity. Enforcing one sampled initial candidate per pair isolates guarded-stage contributions but removes between-condition generation variability. Provider_call_audit and deterministic_reconciliation.json record one repair-stage invocation aligned with the single n_{10} discordant count; thus the observed discordance is attributable to downstream guarded-stage activity recorded in run accounting rather than generation differences.

Operational implications and boundary conditions. For environments where a high-coverage static validator exists and inputs match its expected patterns (parser-validated, constrained intents), adding automated sandboxed emulation may provide negligible aggregate additional detection at scale. This conditional conclusion must not be generalized: in operational regimes with noisier natural-language intents, adversarial inputs, broader vendor diversity, stateful multi-step interactions, or validator gaps, guarded runtime checks and higher-fidelity emulation are likely to provide substantive benefit. The archived per-episode traces support targeted follow-up hypotheses: adversarial task injection, broader vendor-CLI coverage, and complementary validator families should increase the guarded-stage marginal contribution.

VI. LIMITATIONS

- Synthetic task bank and deterministic task generator produce limited ecological diversity and create a validator-ceiling effect; results may not generalize to noisier, adversarial, or production distributions.
- Single declared/requested model family (gpt-5-mini) and single validator implementation (deterministic_netops_validator_v5) limit generalizability across models and validator families.
- Preregistered shared_initial_candidate semantics (one initial generation reused across paired conditions) isolate guarded-stage contribution but remove between-condition generation variability; this design choice constrains dis-

cordance sources to downstream guarded-stage observations or repair calls.

- Sandboxed-emulation fidelity relative to vendor/OS production behaviors and large-scale stateful interactions is limited by the emulation harness used; higher-fidelity emulators could change outcomes in other regimes.
- Study power planning targeted a minimum detectable paired difference of 0.20; the observed aggregate effect (~ 0.0014) lies far below that design sensitivity and should be interpreted as a narrowly scoped near-zero finding for this experimental regime rather than a general negative result for all NetOps workflows.
- Provider-side nondeterminism and hidden caching remain residual uncertainties despite deterministic reconciliation procedures; these are documented in `analysis/deterministic_reconciliation.json` and `provider_call_audit` artifacts.

VII. CONCLUSION

We preregistered and executed a paired confirmatory study comparing a strong deterministic validator with the same validator augmented by sandboxed emulation on a curated synthetic intent→configuration task bank. Over 720 paired tasks the guarded pipeline produced a negligible aggregate detection gain (0.001388888888888884) with exact McNemar $p = 1.0$ and 95% bootstrap CI [0.0, 0.004166666666666667]. Run accounting shows one repair-stage invocation corresponding to the single guarded-only detection; because `shared_initial_candidate` semantics were enforced, archived per-episode traces permit independent verification of that mapping via `execution/raw_results.jsonl` and `analysis/deterministic_reconciliation.json`. We interpret the result as arising from validator ceiling effects and curated task design; follow-up work should focus on adversarial and production-like task banks, multi-validator ensembles, and higher-fidelity emulation to identify regimes where guarded stages add substantive operational value.

Reproducibility pointers (compact). The canonical execution bundle archived by the run contains: `execution/execution_manifest.json` (execution summary and preregistration SHA-256), `execution/raw_results.jsonl` (input-results SHA-256 `df367ecc40a6f0eaf402129260e11d8450884a4ea5fdb94a602e0d2513aef0e8`), `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e049ccf5b53c35ddac2351ad75e7e95504dc4a42bcb096d0dec3aa`), `execution/model_configuration.json` (sha256 `a40e96e212ab87da251ac0d6dbb75bc543afa13438b5a6b9ebd073597ffdd820`), `analysis/paired_contingency_table.csv` (sha256 `efe6e8e7016fa843a8226e911dbde829e943903d719101da8d956b3ff899133c`), and `analysis/condition_summary.csv` (sha256 `394bc690671b731194f9b42b4dcbe28fded6e2ec8cf898b79831225343c22c28`).

The discordant episode is the unique `raw_results.jsonl` entry with `baseline_score = 0` and `guarded_score = 1`; that record contains the `pair_id` and validator/emulation traces that map it to the single repair-stage invocation recorded in

`deterministic_reconciliation.json`. These artifacts are archived as the canonical reproducibility bundle referenced by the execution manifest and the preregistration record. (See Disclosure for exact manifest references.)

DISCLOSURE STATEMENT

Disclosure: Autonomous post-lock research run executed under the frozen master prompt supplied to the system. Declared/requested model: "gpt-5-mini" (provider recorded as `openai_responses`). Deterministic validator: `deterministic_netops_validator_v5`. Orchestration adapter family: `hosted_netops_gvr_v1`. Preregistration id: `f2c3d8b1-7a6e-4a9a-b2c3-5f8e1d2b6c7e` (preregistration SHA-256: `0169919fb8c5aedb2c5a78e031f1ab5c3aeda405cdela80fb2f864768595c1b`; recorded in the execution manifest). Initial master-prompt immutable reference: `provenance/master_prompt.txt`, SHA-256 = `1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba`. Execution summary (artifact-grounded): `completed_episode_count = 1440`, `complete_pair_count = 720`, `model_calls_used = 721` (720 shared initial + 1 repair-stage call); `stage_counts` and `provider-call audit` are archived in `analysis/deterministic_reconciliation.json` (sha256 `49731a2e52e0...`). Human role and autonomy boundary: a human Research Director selected the broad NetOps topic family and constrained the pre-lock master prompt; after the autonomy lock the agentic pipeline autonomously performed literature selection, preregistration, code and prompt generation, experiment execution, analysis, peer review, and manuscript drafting. No manual human editing of technical manuscript content occurred after the autonomy lock. The execution manifest and archived artifacts named above serve as the canonical reproducibility bundle.

REFERENCES

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- [2] Matt Brown, Ari Fogel, Daniel Halperin, Victor Heorhiadi, Ratul Mahajan, Todd Millstein, "Lessons from the evolution of the Batfish configuration analysis tool", 2023, 10.1145/3603269.3604866
- [3] <http://arxiv.org/abs/2407.08249>